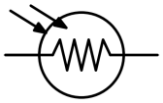
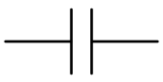
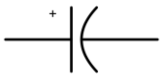
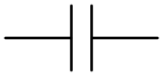
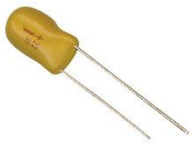
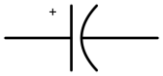
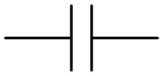
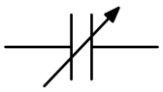
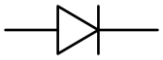
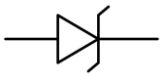
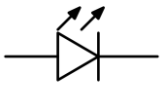
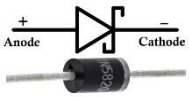
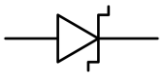
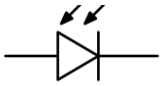
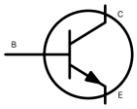
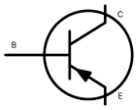
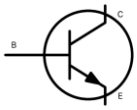
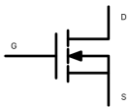
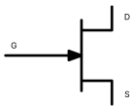
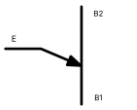
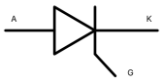
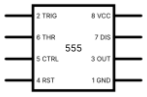
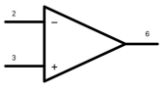
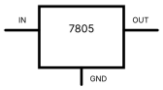
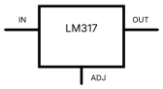
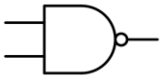
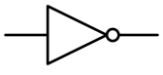
Study of Electronic Components

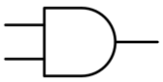
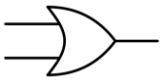
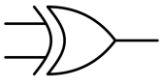
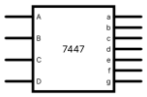
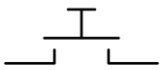
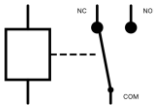
S. No	Components Name	Image	No. of Pins	Pin Details	Application	Symbol
I	**Resistors**					
1.	Carbon Film Resistor		2	Two non-polarized terminals; either lead can be connected to either side.	Current limiting, voltage division, biasing of circuits.	
2.	Metal Film Resistor		2	Two non-polarized terminals; either lead can be connected to either side.	Precision and low-noise circuits, instrumentation, accurate voltage dividers.	
3.	Potentiometer (Variable Resistor)		3	Pin 1 and Pin 3: fixed ends of the resistive track; Pin 2: wiper (variable output).	Volume control, voltage adjustment, sensor calibration.	
4.	Preset (Trimmer) Resistor		3	Pin 1 and Pin 3: fixed ends of the resistive track; Pin 2: wiper (set with a screwdriver).	One-time calibration, bias/offset adjustment, fine tuning of circuits.	
5.	LDR (Light Dependent Resistor)		2	Two non-polarized terminals; resistance decreases as light intensity increases.	Automatic street lights, light meters, dark/light-activated switches, alarms.	
6.	Thermistor (NTC)		2	Two non-polarized terminals; resistance decreases as temperature rises.	Temperature sensing, inrush-current limiting, temperature compensation.	
II	**Capacitors**					
7.	Ceramic Capacitor		2	Two non-polarized terminals; either lead can be connected to either side.	Noise filtering, decoupling, bypass circuits, oscillator circuits.	

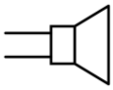
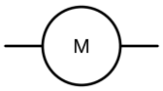
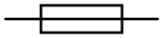
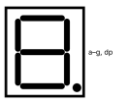
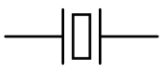
S. No	Components Name	Image	No. of Pins	Pin Details	Application	Symbol
8.	Electrolytic Capacitor		2	Two polarized terminals: Positive (+) (long lead) and Negative (–) (short lead, marked with stripe).	Power supply filtering, smoothing rectifier output, energy storage.	
9.	Mica Capacitor		2	Two non-polarized terminals; either lead can be connected to either side.	High-frequency RF circuits, oscillators, tuned circuits needing stable capacitance.	
10.	Tantalum Capacitor		2	Two polarized terminals: Positive (+) (marked with + sign / longer lead) and Negative (–).	Compact power-supply filtering, decoupling, timing circuits in portable devices.	
11.	Polyester Film Capacitor		2	Two non-polarized terminals; either lead can be connected to either side.	Coupling, decoupling, timing, filters, general-purpose audio and DC circuits.	
12.	Variable (Trimmer) Capacitor		2	Two terminals: fixed plate and rotor (variable) plate; capacitance adjusted with a screwdriver.	Tuning of RF and oscillator circuits, radio receivers, fine calibration.	
III	**Inductors and Transformer**					
13.	Air Core Inductor		2	Two non-polarized terminals (ends of the coil winding).	RF circuits, tuned circuits, high-frequency filters.	
14.	Ferrite Core Inductor		2	Two non-polarized terminals (ends of the coil wound on a ferrite core).	EMI/noise suppression, chokes, SMPS, filter circuits.	

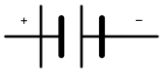
S. No	Components Name	Image	No. of Pins	Pin Details	Application	Symbol
15.	Step-down Transformer		4	Primary: two terminals (mains side, high voltage); Secondary: two terminals (low-voltage output).	Reducing 230 V AC to low AC voltage (e.g., 12 V, 9 V) in power supplies and adapters.	
IV	**Diodes**					
16.	PN Junction Diode (1N4007)		2	Anode (A) and Cathode (K); cathode is marked with a silver band.	Rectification, reverse-polarity protection.	
17.	Zener Diode		2	Anode (A) and Cathode (K); cathode is marked with a band; operated in reverse bias.	Voltage regulation, voltage reference, overvoltage protection, clipping.	
18.	LED		2	Anode (+) (long lead) and Cathode (–) (short lead, flat side of the body).	Indicator lamps, displays, lighting.	
19.	Schottky Diode		2	Anode (A) and Cathode (K); cathode is marked with a band.	High-speed switching, SMPS rectification, low-voltage-drop rectifiers, reverse-polarity protection.	
20.	Photodiode		2	Anode (A) and Cathode (K); operated in reverse bias; cathode is the short lead.	Light detection, optical sensors, IR remote receivers, optical communication.	
21.	Bridge Rectifier		4	Two AC input pins (~) and two DC output pins: Positive (+) and Negative (–).	Full-wave AC-to-DC conversion in power supplies and adapters.	
V	**Transistors**					

S. No	Components Name	Image	No. of Pins	Pin Details	Application	Symbol
22.	BC547 (NPN BJT)		3	Collector (C), Base (B) and Emitter (E); viewed from flat face with leads down: C-B-E.	Switching, amplification, driver circuits.	
23.	BC557 (PNP BJT)		3	Collector (C), Base (B) and Emitter (E); viewed from flat face with leads down: C-B-E.	Switching, amplification, high-side switching, complementary pair with BC547.	
24.	2N2222 (NPN BJT)		3	Emitter (E), Base (B) and Collector (C); in the metal-can package, viewed from below, E is nearest the tab.	General-purpose switching (up to 800 mA), amplifiers, relay and LED drivers.	
25.	IRF540 (N-MOSFET)		3	Gate (G), Drain (D) and Source (S); viewed from the front (text side) with leads down: G-D-S; metal tab is Drain.	High-current switching, motor drivers, SMPS, inverters, PWM dimming.	
26.	JFET (BFW10)		4	Source (S), Drain (D), Gate (G) and Case; N-channel device in a metal can.	RF amplifiers, high-impedance buffers, analog switches, mixers.	
27.	UJT (2N2646)		3	Emitter (E), Base 1 (B1) and Base 2 (B2).	Relaxation oscillators, timing circuits, SCR/TRIAC triggering, sawtooth generation.	
VI	**Thyristors**					
28.	SCR (TYN612)		3	Cathode (K), Anode (A) and Gate (G); viewed from the front with leads down: K-A-G; metal tab is Anode.	Controlled rectifiers, motor speed control, light dimmers, overvoltage protection.	

S. No	Components Name	Image	No. of Pins	Pin Details	Application	Symbol
29.	TRIAC (BT136)		3	Main Terminal 1 (MT1), Main Terminal 2 (MT2) and Gate (G); viewed from the front with leads down: MT1-MT2-G; metal tab is MT2.	AC power control, lamp dimmers, fan regulators, heater controllers.	
VII	**Integrated Circuits**					
30.	IC 555 Timer		8	Pin 1 GND, 2 Trigger, 3 Output, 4 Reset, 5 Control, 6 Threshold, 7 Discharge, 8 VCC.	Timer, astable/monostable multivibrator, pulse generation.	
31.	IC 741 Op-Amp		8	Pin 1 Offset Null, 2 Inverting input (–), 3 Non-inverting input (+), 4 V–, 5 Offset Null, 6 Output, 7 V+, 8 NC.	Amplifiers, comparators, filters, integrators, oscillators.	
32.	IC 7805 Regulator		3	Pin 1 Input, Pin 2 Ground, Pin 3 Output (viewed from the front, leads down).	Fixed +5 V regulated supply for digital circuits and microcontrollers.	
33.	IC LM317 Regulator		3	Pin 1 Adjust (ADJ), Pin 2 Output, Pin 3 Input (viewed from the front, leads down).	Adjustable regulated supply (1.25 V to 37 V), battery chargers, current limiters.	
34.	IC 7400 (NAND)		14	Four 2-input NAND gates; Pin 7 GND, Pin 14 VCC; inputs 1,2 → output 3; 4,5 → 6; 9,10 → 8; 12,13 → 11.	Universal logic gate; building any logic function, latches, oscillators.	
35.	IC 7404 (NOT)		14	Six inverters; Pin 7 GND, Pin 14 VCC; inputs 1, 3, 5, 9, 11, 13 → outputs 2, 4, 6, 8, 10, 12.	Signal inversion, buffering, oscillators, logic circuits.	

S. No	Components Name	Image	No. of Pins	Pin Details	Application	Symbol
36.	IC 7408 (AND)		14	Four 2-input AND gates; Pin 7 GND, Pin 14 VCC; inputs 1,2 → output 3; 4,5 → 6; 9,10 → 8; 12,13 → 11.	Logic AND operation, enable/gating circuits, digital systems.	
37.	IC 7432 (OR)		14	Four 2-input OR gates; Pin 7 GND, Pin 14 VCC; inputs 1,2 → output 3; 4,5 → 6; 9,10 → 8; 12,13 → 11.	Logic OR operation, alarm/combining signals, digital systems.	
38.	IC 7486 (XOR)		14	Four 2-input XOR gates; Pin 7 GND, Pin 14 VCC; inputs 1,2 → output 3; 4,5 → 6; 9,10 → 8; 12,13 → 11.	Adders, parity checkers, comparators, controlled inverters.	
39.	IC 7447 (BCD to 7-Seg Decoder)		16	Inputs A(7), B(1), C(2), D(6); outputs a(13), b(12), c(11), d(10), e(9), f(15), g(14); LT(3), BI/RBO(4), RBI(5); GND 8, VCC 16.	Driving common-anode 7-segment displays from BCD input in counters and clocks.	
VIII	**Switches**					
40.	Push Button Switch		4	Four legs in two internally connected pairs; pressing the button connects the two pairs. Non-polarized.	Manual triggering, reset, input to digital circuits.	
41.	Toggle Switch (SPDT)		3	Common (C) terminal in the middle and two throw terminals (position 1 and position 2).	Selecting between two circuits, ON/OFF and changeover control.	
IX	**Electromechanical Components**					
42.	Relay (5V/12V)		5	Coil 1 and Coil 2 (coil terminals); Common (COM), Normally Open (NO) and Normally Closed (NC) contacts.	Switching high-power loads with a low-power signal, isolation, automation.	

S. No	Components Name	Image	No. of Pins	Pin Details	Application	Symbol
43.	Buzzer		2	Two polarized terminals: Positive (+) (long lead / marked) and Negative (–).	Alarms, indicators, timers, notification sounds.	
44.	Speaker (8 Ω)		2	Two terminals (voice coil): Positive (+) and Negative (–); 8 Ω impedance.	Audio output in amplifiers, radios, alarms and music players.	
45.	DC Motor		2	Two terminals: Positive (+) and Negative (–); reversing polarity reverses the direction of rotation.	Fans, toys, robotics, pumps, small drives.	
X	**Protection**					
46.	Fuse		2	Two non-polarized terminals; either lead can be connected to either side.	Overcurrent protection; the element melts and opens the circuit when current exceeds its rating.	
XI	**Display**					
47.	7-Segment Display		10	Segment pins a, b, c, d, e, f, g and DP, plus common pin(s) (COM): common cathode or common anode.	Displaying numeric digits in clocks, counters, meters and calculators.	
XII	**Oscillator**					
48.	Crystal Oscillator		2	Two non-polarized terminals (quartz crystal in a metal can); either lead can be connected to either side.	Clock source for microcontrollers, frequency generation, timers, communication circuits.	
XIII	**Source**					

S. No	Components Name	Image	No. of Pins	Pin Details	Application	Symbol
49.	Battery / DC Cell		2	Positive (+) terminal (longer line in the symbol) and Negative (–) terminal.	DC power source for portable and low-voltage circuits.	
50.	Optocoupler (PC817)		4	Pin 1: Anode and Pin 2: Cathode (these form the input LED side); Pin 3: Emitter and Pin 4: Collector (these form the output phototransistor side)	Provides complete electrical isolation between two circuits while allowing signal transmission, protecting sensitive components from voltage spikes	